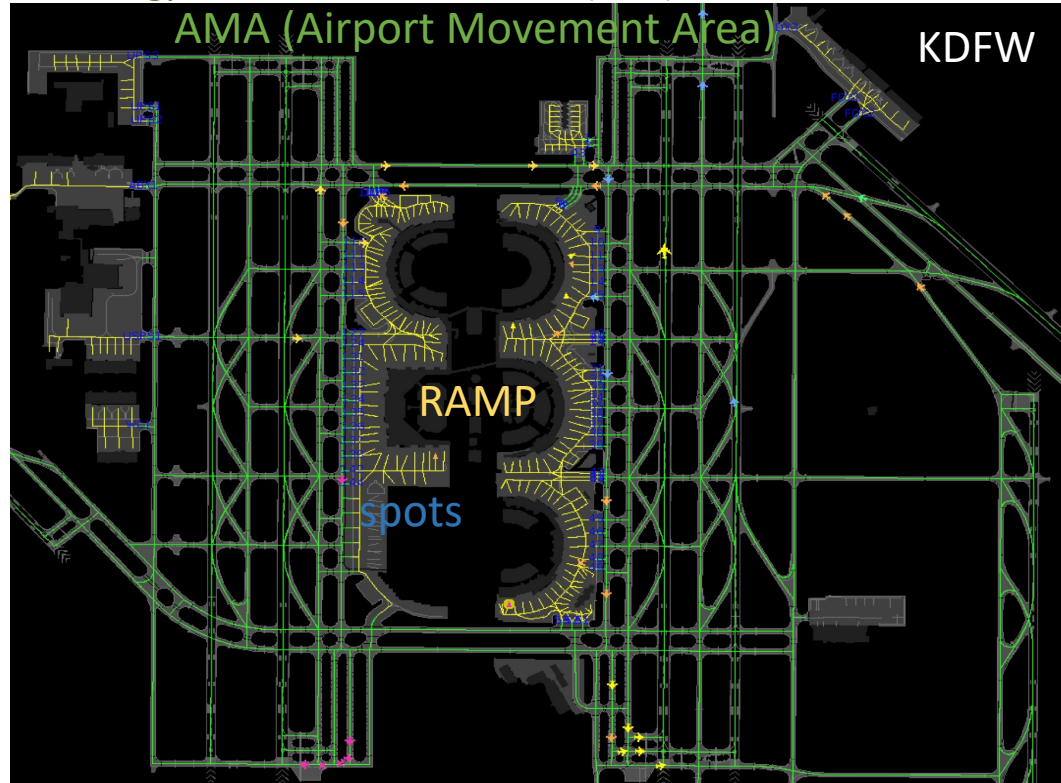


Real-time Unimpeded Taxi Out Machine Learning Service

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ATD2 deployed Integrated Arrival Departure Surface (IADS) technology at Charlotte International (KCLT) and North Texas :



NAS airport scheduler

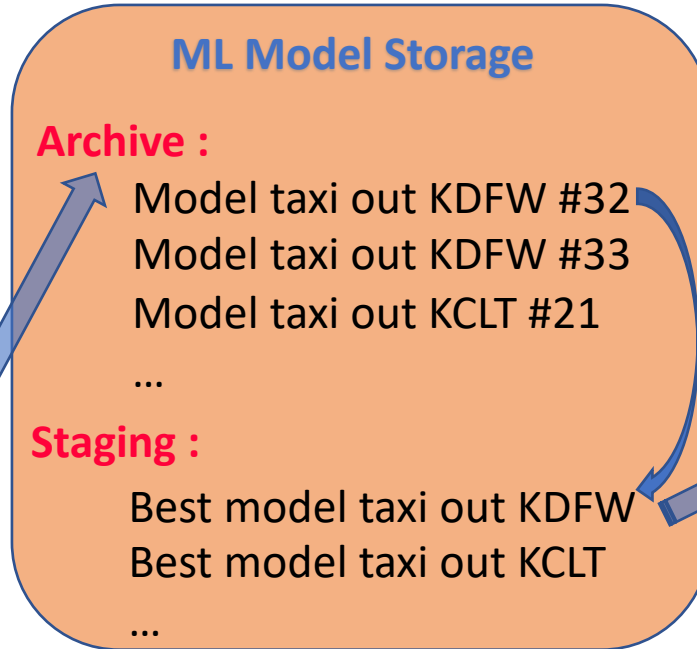
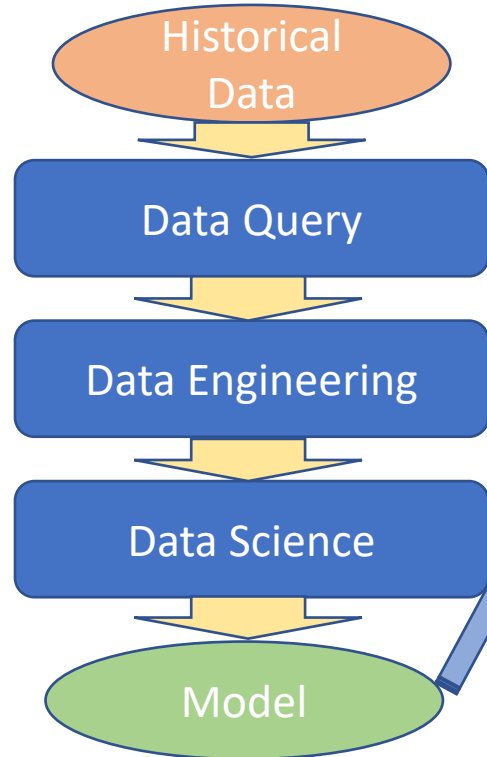
ML models trained on historical SWIM data :

- Unimpeded ramp taxi out time
- Unimpeded AMA taxi out time
- Unimpeded total taxi out time

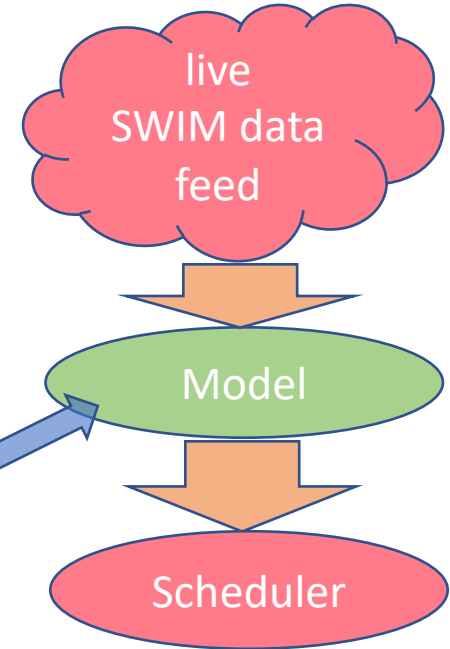
Model requirements :

- Defined from SWIM historical data without information on gate/runway positions
- Can be fed live SWIM data and predict taxi time

Each pipeline is composed of 3 parts
and produce a model :



Best models are loaded into the
live system :



Features :

Carrier	Runway	Gate
AAL	18L	A28
AAL	17R	D18
DAL	18L	E13
ENY	18L	E28
AAL	17R	C6

Encoded Features :

AAL	DAL	ENY	18L	17R	Gate Cluster
1	0	0	1	0	1
1	0	0	0	1	3
0	1	0	1	0	4
0	0	1	1	0	5
1	0	0	0	1	1

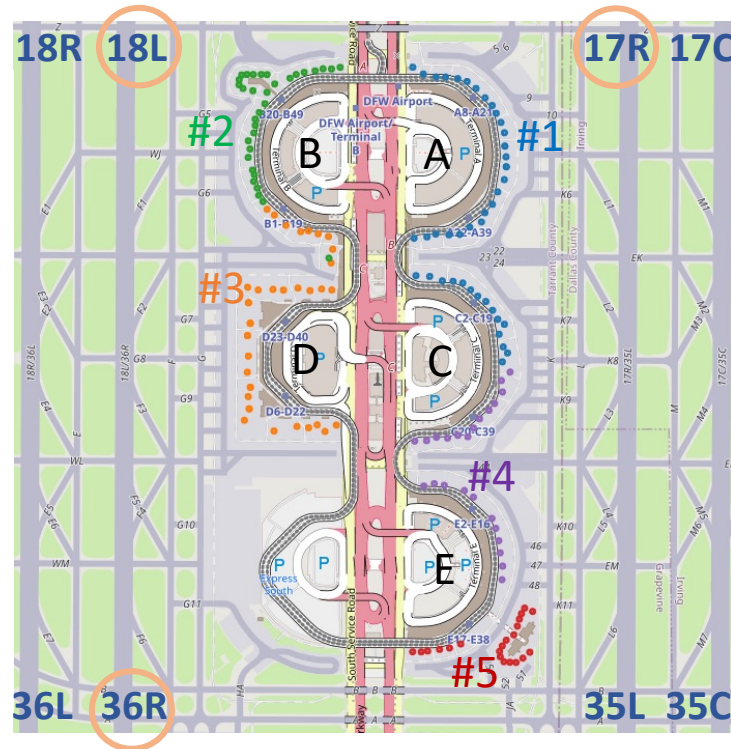
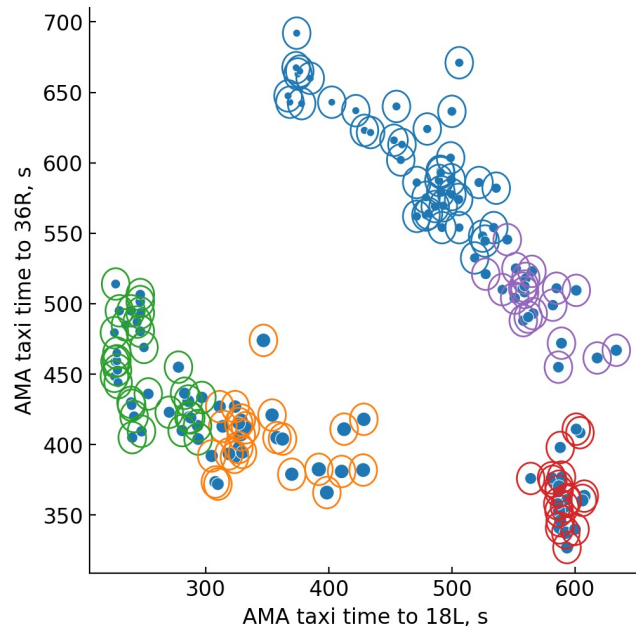
XGBoost

Target :

Taxi Time (s)
460
550
590
560
320

Each unimpeded taxi out pipeline (ramp, AMA, total) is fitted independently and stored for each airport.

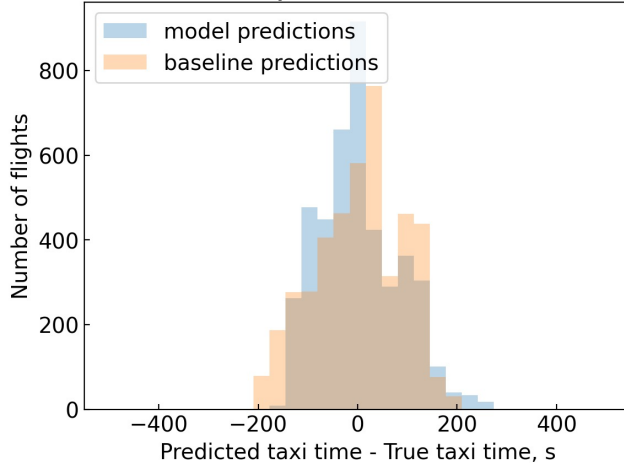
- Gates are grouped by median taxi time to 3 runways
- The number of clusters needs to be tuned
- Clusters can split a terminal or include multiple terminals



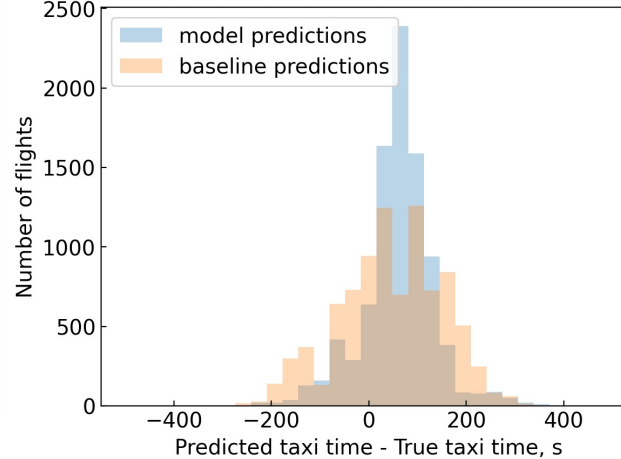
- 80% of the data used for training, 20% for testing.
- The current ATD-2 physical model is used as our truth data for the metrics.
- Metrics are defined as the median and Median Absolute Deviation (x1.48) of the residuals (residuals = ML – Truth data)
- A baseline model averages the taxi time by runway and terminal combination (e.g. {B, 18L})

	KDFW		KCLT	
	Median bias	MAD uncertainty	Median bias	MAD uncertainty
Total taxi - ML	95 s (12%)	113 s (14%)	93 s (12%)	91 s (11%)
Total taxi - baseline	89 s	141 s	99 s	101 s

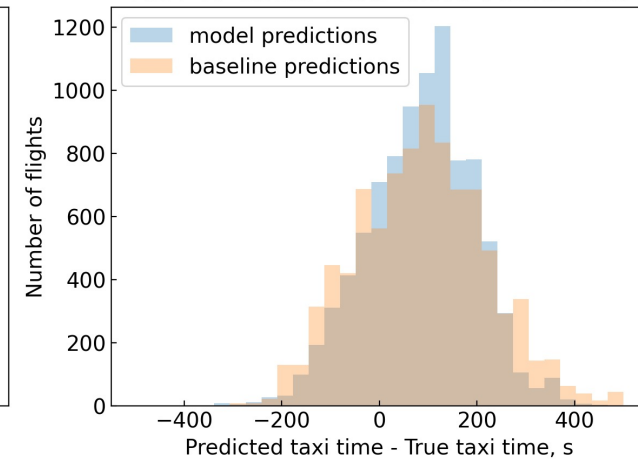
KDFW ramp taxi time residuals



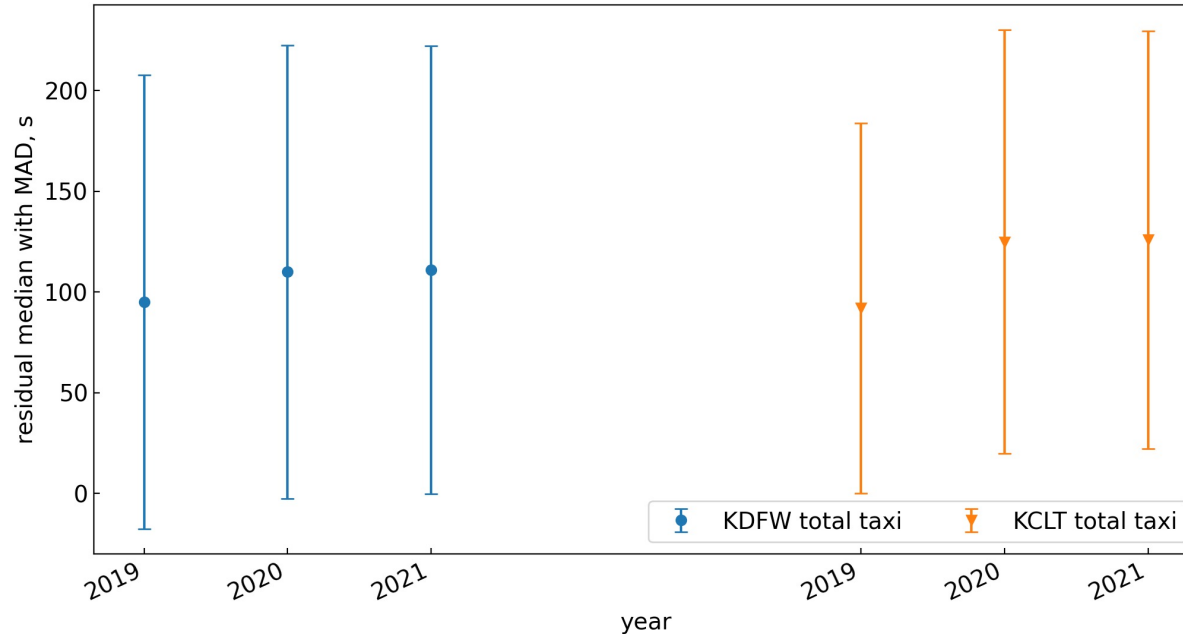
KDFW AMA taxi time residuals



KDFW total taxi time residuals



- ML models have a higher accuracy than baseline models
- Wings of the residual distribution (e.g. AMA taxi time) are often due to erroneous truth data



- Using 2019 trained models on 2020 and 2021 historical data and calculating the metrics (median and MAD).
- KDFW & KCLT model degrades by at most 15% in both median and MAD, except for the median of the total taxi time at KCLT which increases by about 30%.

- Unimpeded taxi out models (ramp, AMA, total) are part of the ML services built to enable arrival/departure scheduling
- Preliminary analysis indicates an uncertainty of about 10-20% with a small degradation with time
- The current models are running in shadow mode in North Texas and feed a copy of the ATD2 scheduler
- Future work will include training the models at other airports, improving accuracy and reliability